



مدارس
للتكنولوجيا
التطبيقية

IT Department
Web Programming

Practical Exercises
الصف الثالث

Practical Exercises

2st

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Practical Exercises

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Please be advised that the materials contained in this book have been specifically developed to reflect the most current information, trends, and standards in the field.

It is critical to your success that you rely solely on the content provided here and refrain from using any outdated resources or materials from previous years.



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Exercise	1
Name	PHP1
Outcomes and Performance Criteria	➤ Develop a functional PHP application that demonstrates the use of variables, functions, and built-in functions. ➤ Build a complete PHP project that includes handling arrays, loops, conditionals, and form data processing. ➤ Create a set of PHP scripts that integrate with HTML forms ➤ Construct a small web application using PHP.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	45 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Exercise Scenario: Create Student Report Portal

The school needs a simple web page to view a list of students and their performance. Your task is to build the core PHP logic and the view for this portal.

Instructions:

1. **Data Setup:** In your PHP script, model the student data by creating a multi-dimensional array named `$students`. It should hold records for 5 students, with each record including the student's **Name**, **Grade**, **Subject**, and **Marks**.
2. **Data Loop:** Write a PHP foreach loop to access each student's record from the array.
3. **Visual Presentation:** Format the output within the loop to display the data in a clear and organized **HTML table**. Apply styling using Bootstrap to enhance the user interface (e.g., add alternating row colors, borders, and header styling).

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Exercise	2
Name	Advanced PHP 1
Outcomes and Performance Criteria	➤ Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. ➤ Build a functional CRUD application with user authentication and error handling.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	45 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Exercise Scenario: Create a 3-page website for "Cambridge International College" using PHP includes for consistent header, navigation, and footer.

Requirements:

1. Create modular files: header.php, navigation.php, footer.php
2. Build 3 pages using these components:
 - index.php - Home page with introduction
 - courses.php - Courses listing
 - contact.php - Contact information
3. Use PHP include/require to assemble pages
4. Ensure consistent navigation across all pages

Deliverable: Fully functional website with reusable components and seamless navigation.

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Exercise	3
Name	Advanced PHP 1
Outcomes and Performance Criteria	➤ Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. ➤ Build a functional CRUD application with user authentication and error handling.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	45 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Exercise Scenario: Build a website for "La Bella Trattoria" restaurant with consistent styling across pages.

Requirements:

1. Create modular files:
 - head.php - HTML head section
 - navbar.php - Restaurant navigation
 - footer.php - Address and social links
2. Develop 3 pages:
 - home.php - Welcome and featured dishes
 - menu.php - Full restaurant menu
 - location.php - Address and opening hours
3. Use PHP includes to assemble pages
4. Ensure seamless navigation between pages

Deliverable: Complete restaurant website with consistent layout and fully functional navigation.

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Exercise	4
Name	Advanced PHP 1
Outcomes and Performance Criteria	➤ Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. ➤ Build a functional CRUD application with user authentication and error handling.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	60 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Calculator Exercise

Number 1

Number 2

Operation

☒ Add ☐ Subtract ☐ Multiply ☐ Divide

Reset

Calculate

Exercise Scenario: Develop a web-based calculator for "EduTech Solutions" learning platform using Bootstrap and PHP.

Requirements:

1. Design: Create a clean form using Bootstrap with:
 - Two number input fields
 - Operation dropdown (Add, Subtract, Multiply, Divide)
 - Calculate and Reset buttons
2. Functionality:
 - Use POST method to submit data
 - Process calculations on same page
 - Display results clearly
 - Handle errors (e.g., division by zero)
3. Features:
 - Responsive, user-friendly design
 - Clear form labels and layout
 - Proper result display

Deliverable: Single PHP file with form interface and calculation logic.

Exercise	5
Name	Advanced PHP 2
Outcomes and Performance Criteria	➤ Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. ➤ Build a functional CRUD application with user authentication and error handling.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	120 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Exercise Scenario: Build a web application for "City College" to manage student records using PDO and Bootstrap.

Requirements:

1. Database:

- Create college_records database with students table
- Include fields: id, name, email, phone, course

2. PDO Operations:

- Secure database connection using PDO
- Implement Create, Read, Delete functionality

3. Features:

- **Add Students:** Form with validation
- **View Students:** Display all records in table
- **Delete Students:** Remove records with confirmation

4. Interface:

- Use Bootstrap for responsive design
- Professional forms and tables
- User-friendly feedback messages

Deliverable: Complete web application with secure database operations and polished Bootstrap interface.

Exercise	6
Name	Advanced PHP 2
Outcomes and Performance Criteria	➤ Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. ➤ Build a functional CRUD application with user authentication and error handling.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	45 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Exercise Scenario: Build a language preference system for "GlobalConnect Inc." using PHP cookies.

Requirements:

1. Language Selection:

- Dropdown form with Arabic, English, French options
- Submit button to save preference

2. Cookie Management:

- Store selection in "user_language" cookie (365 days expiry)
- Set "english" as default if no cookie exists

3. User Interface:

- Display current language preference
- Show confirmation message when preference is saved
- Clear, user-friendly design

Deliverable: PHP application that remembers user language choices and provides seamless multilingual support.

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Exercise	7
Name	Advanced PHP 2
Outcomes and Performance Criteria	➤ Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. ➤ Build a functional CRUD application with user authentication and error handling.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	120 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Scenario: Build an inventory management system for "RetailPro Store" using PDO and Bootstrap.

Requirements:

1. Database:

- Create retailpro_inventory database with products table
- Fields: product_id, product_name, category, price, stock_quantity, supplier

2. PDO Operations:

- Secure PDO database connection
- Implement Create, Read, Delete functionality

3. Features:

- **Add Products:** Form with product details and validation
- **View Products:** Display all inventory items with stock status
- **Delete Products:** Remove discontinued items with confirmation

4. Interface:

- Bootstrap for clean, responsive design
- Professional product catalog display
- Low stock alerts and category filtering

Deliverable: Complete inventory management application with secure database operations and professional interface.

Exercise	8
Name	Advanced PHP 2
Outcomes and Performance Criteria	➤ Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. ➤ Build a functional CRUD application with user authentication and error handling.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms
Time Required	45 mins
Tools needed	VS Code Xampp Bootstrap Web Browser

Exercise Scenario: Build a visitor tracking system for "Digital Insights Agency" using PHP cookies.

Requirements:

1. Cookie Tracking:

- Check for 'visit_count' cookie
- Initialize to 1 for new visitors
- Increment by 1 for returning visitors
- Set 30-day expiration

2. Functionality:

- Maintain accurate count across sessions
- Handle cookie creation and updates

3. User Interface:

- Display current visit count prominently
- Show personalized messages for new/returning visitors
- Clean, professional design

Deliverable: PHP application that reliably tracks and displays website visit counts using cookies.

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Exercise	9
Name	Laravel 1
Outcomes and Performance Criteria	➤ Develop complete web applications using Laravel, demonstrating the integration of routes, controllers, models, and views. ➤ Build dynamic web pages using Blade templates with layout inheritance, components, and conditional logic. ➤ Create and manage database tables using migrations and implement RESTful APIs with Laravel to handle CRUD operations.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms TPC5.6: Storing and using data by creating data tables and raw SQL statements TPC5.7 : Using controllers and routes for APIs and URLs TPC5.8: Creating and using composer packages TPC5.9: Create restful services, use an Effect Dependency Array and how to hand errors in data requests
Time Required	90 mins
Tools needed	VS Code Xampp Composer Node.js Web Browser Bootstrap Font Awesome

Exercise Scenario: Build a movie collection manager for "Cinephile Hub" using Laravel.

Requirements:

1. Database & Models:

- Create movies migration with: id, title, director, release_year, rating
- Create Movie Eloquent model
- Implement MovieSeeder with sample data

2. Controllers & Routes:

- Create MovieController with resource methods
- Set up routes for movie listing and detail views

3. Blade Views:

- movies.index - Bootstrap grid layout
- movies.show - Individual movie page
- Responsive, professional design

4. Features:

- Movie catalog with main listing
- Detailed movie pages
- Clean navigation
- Bootstrap styling

Deliverable: Laravel MVC application with complete movie library functionality.

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Exercise	10
Name	Laravel 2
Outcomes and Performance Criteria	➤ Develop complete web applications using Laravel, demonstrating the integration of routes, controllers, models, and views. ➤ Build dynamic web pages using Blade templates with layout inheritance, components, and conditional logic. ➤ Create and manage database tables using migrations and implement RESTful APIs with Laravel to handle CRUD operations.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms TPC5.6: Storing and using data by creating data tables and raw SQL statements TPC5.7 : Using controllers and routes for APIs and URLs TPC5.8: Creating and using composer packages TPC5.9: Create restful services, use an Effect Dependency Array and how to hand errors in data requests
Time Required	90 mins
Tools needed	VS Code Xampp

Composer Node.js Web Browser Bootstrap Font Awesome

Exercise Scenario: Build a product management API for "E-Commerce Solutions Inc." using Laravel.

Requirements:

1. Setup:

- New Laravel project with database
- API routing configuration

2. Product System:

- Model, migration, factory, seeder
- Fields: name, description, price, category, stock

3. API Endpoints:

- Full CRUD operations (GET, POST, PUT/PATCH, DELETE)
- Product listing, creation, details, update, deletion

4. Testing:

- Thorough Postman testing
- Validate all CRUD operations

Deliverable: Fully functional Laravel API with complete product management capabilities.

Exercise	11
Name	Laravel 3
Outcomes and Performance Criteria	➤ Develop complete web applications using Laravel, demonstrating the integration of routes, controllers, models, and views. ➤ Build dynamic web pages using Blade templates with layout inheritance, components, and conditional logic. ➤ Create and manage database tables using migrations and implement RESTful APIs with Laravel to handle CRUD operations.
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.1 : Use database software tools to run queries and produce reports TPC5.2: Write PHP scripts to handle HTML forms TPC5.3: Develop and use simple Dashboards and simple data visualization forms TPC5.6: Storing and using data by creating data tables and raw SQL statements TPC5.7 : Using controllers and routes for APIs and URLs TPC5.8: Creating and using composer packages TPC5.9: Create restful services, use an Effect Dependency Array and how to hand errors in data requests
Time Required	90 mins
Tools needed	VS Code Xampp

Composer Node.js Web Browser Bootstrap Font Awesome

Exercise Scenario: Build a blog management API for "Digital Publishers Ltd." using Laravel.

Requirements:

1. Setup:

- New Laravel project with database
- API routing configuration

2. Blog System:

- Post model, migration, factory, seeder
- Fields: title, content, author, category, published_at, status

3. API Endpoints:

- Full CRUD operations for blog posts
- Create, read, update, delete functionality

4. Testing:

- Comprehensive Postman testing
- Validate all blog management operations

Deliverable: Complete Laravel blog API with full content management capabilities.

Exercise	12
Name	React 1
Outcomes and Performance Criteria	➤ Build a dynamic React application that utilizes components, state, props, and Hooks for interactivity and state management. ➤ Design Simple component hierarchies with effective state sharing using useState. ➤
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.4: Using UI FrameWork (React)
Time Required	30 mins
Tools needed	VS Code Internet React Project Web Browser Node.js Bootstrap

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Exercise Scenario: Build an interactive counter for "WebTech Solutions" using React and Bootstrap.

Requirements:

1. State:

- Use useState hook for counter state
- Implement setCounter function

2. Functions:

- increment: +1 to counter
- decrement: -1 to counter
- reset: set to 0

3. UI:

- Three Bootstrap buttons (Increment, Decrement, Reset)
- Prominent counter display
- Clean, responsive design

Deliverable: Functional React counter application with state management and Bootstrap styling.

Exercise	13
Name	React 2
Outcomes and Performance Criteria	➤ Build a dynamic React application that utilizes components, state, props, and Hooks for interactivity and state management. ➤ Design Simple component hierarchies with effective state sharing using useState. ➤
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.4: Using UI FrameWork (React)
Time Required	45 mins
Tools needed	VS Code Internet React Project Web Browser Node.js Bootstrap Images Folder on the Desktop

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Exercise Scenario: Build a portfolio showcase for "DesignStudio Pro" using React and Bootstrap.

Requirements:

1. Setup:

- React app with Bootstrap
- Public folder for images

2. Data:

- Create array of card objects
- Include id, image, title, description, buttonText

3. Card Component:

- Reusable component with props
- Accepts image, title, description, buttonText
- Bootstrap styling

4. Rendering:

- Use `map()` to iterate through array
- Render multiple cards dynamically
- Responsive grid layout

Deliverable: React app using array and `map()` to render multiple reusable card components with dynamic content.

Exercise	14
Name	React 3
Outcomes and Performance Criteria	➤ Build a dynamic React application that utilizes components, state, props, and Hooks for interactivity and state management. ➤ Design Simple component hierarchies with effective state sharing using useState. ➤
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.4: Using UI FrameWork (React)
Time Required	45 mins
Tools needed	VS Code Internet React Project Web Browser Node.js Bootstrap

Exercise Scenario: Build an interactive like button for "ConnectSocial App" using React useState.

Requirements:

1. State:

- Use useState for likes count
- Initialize to 0

2. Functionality:

- handleLike function to increment count
- Real-time updates on click

3. UI:

- Bootstrap-styled like button
- Display likes count
- Visual feedback

Deliverable: React like button component with state management and counter functionality.

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Exercise	15
Name	React 4
Outcomes and Performance Criteria	➤ Build a dynamic React application that utilizes components, state, props, and Hooks for interactivity and state management. ➤ Design Simple component hierarchies with effective state sharing using useState. ➤
Skills	TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.5: Control applications layout with geometry managers TPC4.6: Create interactive applications TPC4.7: Create native design TPC4.8: Create UI web design TPC5.4: Using UI FrameWork (React)
Time Required	45 mins
Tools needed	VS Code Internet React Project Web Browser Node.js Bootstrap

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Exercise Scenario: Build a theme switcher for "TechInnovate Solutions" using React useState.

Requirements:

1. State:

- Use useState for isDarkMode
- Initialize to false

2. Functionality:

- Toggle between light/dark themes
- Dynamic background and text colors

3. UI:

- Button with dynamic text
- Visual theme feedback
- Accessible color contrast

Deliverable: React theme toggler component with state-managed light/dark mode switching.

Exercise	16
Name	Testing 1
Outcomes and Performance Criteria	➤ Understand how to interact with APIs using Postman. ➤ Verify endpoints for CRUD operations. ➤ Learn to validate response details using tests in Postman.
Skills	TPC3.1: Use a APIs TPC3.3: Secure an application TPC4.1: Create and maintain websites
Time Required	45 mins
Tools needed	VS Code Internet Web Browser Postman

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Exercise Scenario: Build a testing demonstration for "CodeCraft Solutions" using manual unit tests.

Requirements:

1. Function:

- Create add function in mathOperations.js
- Handle two parameters and return sum

2. Testing:

- Create testMathOperations.js
- Test various number combinations
- Compare actual vs. expected results

3. Reporting:

- Log "PASS" or "FAIL" for each test
- Clear test case descriptions
- Comprehensive test coverage

Deliverable: JavaScript function with manual test script demonstrating unit testing fundamentals.

Exercise	17
Name	Testing 2
Outcomes and Performance Criteria	➤ Build a dynamic JavaScript application that utilizes Testing ➤ Discover bugs in a software Development . ➤ Understand Testing In Software
Skills	TPC3.3: Secure an application TPC3.4: Discover bugs in a software TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.6: Create interactive applications
Time Required	45 mins
Tools needed	VS Code Internet Web Browser

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Exercise Scenario: Implement professional testing for "TechQuality Assurance" using Jest framework.

Requirements:

1. Setup:

- Install and configure Jest
- Create project structure

2. Function:

- Implement add function in add.js
- Export for testing

3. Testing:

- Create add.test.js with multiple test cases
- Test various number combinations
- Use Jest assertions

4. Execution:

- Run tests with `npx jest`
- Verify all tests pass

Deliverable: Jest testing setup with comprehensive test suite for addition function.

Exercise	18
Name	Testing 3
Outcomes and Performance Criteria	➤ Discover bugs in a software Development. ➤ Understand Testing in Software
Skills	TPC3.3: Secure an application TPC3.4: Discover bugs in a software TPC4.1: Create and maintain websites TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.6: Create interactive applications
Time Required	60 mins
Tools needed	VS Code Internet Web Browser Node.js Jest-Dom Bootstrap

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Exercise Scenario: Build integration tests for "FinTech Solutions Inc." calculator without external libraries.

Requirements:

1. Calculator Module:

- Create calculator.js with add, subtract, multiply, divide
- Include error handling

2. Integration Tests:

- Create calculator.test.js
- Test function combinations
- Simulate real user workflows

3. Test Execution:

- Manual test runner
- Clear pass/fail reporting
- Error handling

Deliverable: Integration test suite verifying calculator functions work correctly together.

Exercise	19
Name	Testing 4
Outcomes and Performance Criteria	➤ Understand how to interact with APIs using Postman. ➤ Verify endpoints for CRUD operations. ➤ Learn to Testing Performance of The Website Using Jmeter.
Skills	TPC3.1: Use a APIs TPC3.3: Secure an application TPC4.1: Create and maintain websites TPC4.4: Use basic data storage software and Digital Design Software TPC4.6: Create interactive applications
Time Required	60 mins
Tools needed	VS Code Internet Web Browser Jmeter

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Exercise Scenario: Build integration tests for "Enterprise Finance Corp" using Jest framework.

Requirements:

1. Calculator Module:

- Create calculator.js with arithmetic functions
- Include error handling

2. Jest Tests:

- Create calculator.test.js
- Test function combinations
- Use Jest assertions

3. Execution:

- Run tests with npx jest
- Verify integration scenarios

Deliverable: Jest integration test suite validating calculator function interactions.

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Exercise	20
Name	Node.js 1
Outcomes and Performance Criteria	➤ Build a dynamic React application that utilizes components, state, props, and Hooks for interactivity and state management. ➤ Discover bugs in a software Development . ➤ Understand Testing In Software
Skills	TPC3.3: Use Apis TPC4.1: Create and maintain websites TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.6: Create interactive applications TPC5.1: Use database software tools to run queries and produce reports TPC5.9: Create restful services, use an Effect Dependency Array and how to hand errors in data requests
Time Required	90 mins
Tools needed	VS Code Internet Web Browser Node.js Project Postman Mysql Nodemon Express.js

Exercise Scenario: Build a product management API for "Global Retail Solutions" using Node.js and MySQL.

Requirements:

1. Setup:

- Node.js with Express, MySQL, Nodemon
- we-db database with products table

2. API Endpoints:

- POST /products - Create product
- GET /products - Read all products
- GET /products/:id - Read single product
- PUT /products/:id - Update product
- DELETE /products/:id - Delete product

3. Testing:

- Verify all CRUD operations
- Test API functionality

Deliverable: Fully functional Node.js REST API for product management.

Exercise	21
Name	Node.js 2
Outcomes and Performance Criteria	➤ Understand how to interact with APIs using Postman. ➤ Verify endpoints for CRUD operations. ➤ Learn to validate response details using tests in Postman.
Skills	TPC3.1: Use a APIs TPC3.3: Secure an application TPC4.1: Create and maintain websites
Time Required	45 mins
Tools needed	VS Code Internet Web Browser Node.js App Express.js

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Exercise Scenario: Build a corporate website for "Innovate Web Solutions" using Express.js.

Requirements:

1. Setup:

- Install Express
- Create index.js

2. Routes:

- Home (/)
- About (/about)
- Contact (/contact)
- Products (/products)

3. Functionality:

- Render appropriate pages
- Test all routes
- Ensure application runs correctly

Deliverable: Multi-page Express application with working routes and professional content.

Exercise	22
Name	Design Pattern 1
Outcomes and Performance Criteria	➤ Build a dynamic React application that utilizes components, state, props, and Hooks for interactivity and state management. ➤ Discover bugs in a software Development . ➤ Understand Testing In Software
Skills	TPC3.4: Discover bugs in a software TPC4.1: Create and maintain websites TPC4.2: Access bootstrap Plug-Ins TPC4.3: Dynamically add content to a web page using JavaScript TPC4.4: Use basic data storage software and Digital Design Software TPC4.6: Create interactive applications TPC4.8: Create UI web design TPC5.2: Write PHP scripts to handle HTML forms TPC5.4: Using UI FrameWork (React) TPC5.9: Create restful services, use an Effect Dependency Array and how to hand errors in data requests
Time Required	45 mins
Tools needed	VS Code Internet Web Browser Node.js Project Postman Mysql Nodemon Express.js

Exercise Scenario: Build a car manufacturing system for "AutoTech Solutions" using Factory Design Pattern.

Requirements:

1. Car Class:

- Create Car class with type and model

2. Factory:

- Implement CarFactory with createCar method
- Return new Car instances

3. Demonstration:

- Create multiple car objects
- Test with different types and models

Deliverable: Factory Pattern implementation for dynamic car object creation.

**PRACTICE
MAKES
progress**

Exercise	23
Name	Design Pattern 2
Outcomes and Performance Criteria	➤ Understand how to interact with APIs using Postman. ➤ Verify endpoints for CRUD operations. ➤ Learn to validate response details using tests in Postman.
Skills	TPC3.1: Use a APIs TPC3.3: Secure an application TPC4.1: Create and maintain websites
Time Required	45 mins
Tools needed	VS Code Internet Web Browser Node.js App Express.js
Expected Result	Expected result in Console: Preparing Tea: Boiling water... Steeping the tea... Pouring into cup... Adding lemon... Preparing Coffee: Boiling water... Dripping coffee through filter... Pouring into cup... Adding sugar and milk...

Exercise Scenario: Integrate legacy printing system with modern framework for "TechModern Solutions".

Requirements:

1. Legacy System:

- LegacyPrinter with printLegacy() method

2. Modern Interface:

- Printer class with print() method

3. Adapter:

- PrinterAdapter extending Printer
- Adapt printLegacy() to print()

4. Integration:

- Demonstrate seamless usage
- Preserve legacy functionality

Deliverable: Adapter Pattern implementation bridging legacy and modern printing systems.

Exercise	24
Name	Advanced Python 2
Outcomes and Performance Criteria	➤ Design and develop a complete Tkinter application that includes layout management, widget customization, and user input handling. ➤ Build a multi-widget application using text widgets for displaying and processing user data. ➤ Combine various Tkinter components creatively to create visually appealing and functional GUIs.
Skills	TPC4.1: Create and maintain websites TPC4.4: Use basic data storage software and Digital Design Software TPC4.6: Create interactive applications TPC5.9: Create restful services, use an Effect Dependency Array and how to hand errors in data requests TPC6.2: Solving problems using algorithms TPC6.5: Handle and produce errors (builtin or custom) to process or signal failure
Time Required	45 mins
Tools needed	VS Code Internet Web Browser Pycharm

**PRACTICE
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Exercise Scenario: Build a secure login interface for "SecureAccess Corp" using Python Tkinter.

Requirements:

1. Setup:

- Create login_form.py with Tkinter
- Configure main application window

2. GUI Components:

- Username and password labels
- Input entry fields
- Login button

3. Functionality:

- Login validation logic
- User feedback system
- Event handling

4. Testing:

- Verify form works correctly
- Test validation and feedback

Deliverable: Fully functional Python Tkinter login form application.

Exercise	25
Name	Advanced Python 2
Outcomes and Performance Criteria	➤ Design and develop a complete Tkinter application that includes layout management, widget customization, and user input handling. ➤ Build a multi-widget application using text widgets for displaying and processing user data. ➤ Combine various Tkinter components creatively to create visually appealing and functional GUIs.
Skills	TPC4.1: Create and maintain websites TPC4.4: Use basic data storage software and Digital Design Software TPC4.6: Create interactive applications TPC5.9: Create restful services, use an Effect Dependency Array and how to hand errors in data requests TPC6.2: Solving problems using algorithms TPC6.5: Handle and produce errors (builtin or custom) to process or signal failure
Time Required	60 mins
Tools needed	VS Code Internet Web Browser Pycharm

**PRACTICE
MAKES
progress**

Exercise Scenario: Build a price calculator for "QuickMart Stores" using Python Tkinter.

Requirements:

1. Setup:

- Tkinter application with proper window configuration

2. Interface:

- Price and discount input fields
- Calculate button
- Net price display area

3. Functionality:

- Discount calculation logic
- Input validation
- Real-time results

4. Testing:

- Verify accurate calculations
- Test error handling

Deliverable: Python Tkinter application that calculates net prices after discounts.